

Problem Framing Document

Project: Sentinel-FinAI | Lead-Level Banking Advisory Agent

1. Problem Statement

Modern banking customers often struggle to navigate complex product offerings, interest rates, and bank policies through traditional static websites. Customer support centers are frequently overwhelmed with non-transactional inquiries, leading to long wait times and inconsistent information delivery.

2. Project Objectives

- **Automated Advisory:** Provide a 24/7 intelligent interface that accurately retrieves and explains banking products using RAG (Retrieval-Augmented Generation).
- **Safety & Compliance:** Implement strict boundaries to prevent the AI from providing unauthorized legal, tax, or investment advice.
- **Personalized Experience:** Utilize short-term memory to recognize user identity and maintain conversation context.
- **Operational Efficiency:** Reduce the load on human agents by resolving high-frequency informational queries with high precision.

3. Target Audience

The primary users are retail banking customers seeking immediate, accurate information regarding savings accounts, loan calculations, and eligibility criteria without requiring human intervention.

4. Proposed Solution

Sentinel-FinAI is a multi-modal AI agent built on a LangChain/LangGraph framework. It integrates a vector database (ChromaDB) for specialized knowledge retrieval and custom Pydantic-based tools for financial calculations. The system utilizes an Adaptive Prompt Registry to manage system behaviors and safety protocols across different deployment versions.

5. Success Metrics

- **RAG Accuracy:** Zero hallucinations regarding bank-specific interest rates and policies.
- **Safety Compliance:** 100% refusal rate for out-of-scope financial advice queries.
- **System Latency:** Average response time maintained under 3 seconds for optimal user experience.